balancing, chassis design, potential obstructions, and more. Our machine found a way to provide 96% vehicle coverage using only 15 sensors. So, when you ask about potential, we're talking about saving autonomous vehicle designers a tremendous amount of money. This is compared to the solutions provided by other quantum and classical computers that suggested using well over 100 sensors," explains Liscouski.

Mercedes-Benz, which hopes for all its vehicles to be carbon-neutral by 2039, is working with IBM Quantum to simulate the chemical reactions in batteries more accurately. Basically, they are trying to understand what happens at a molecular level inside the battery, while it is working. This is a task that is practically impossible for classical supercomputers, as it involves a tremendous number of electron interactions, each influencing the other in complex ways. No wonder, Dr Jeannette (Jamie) Garcia, Senior Manager of Quantum Algorithms, Applications and Theory at IBM, remarks that, "It is chaos in there!"

"Most industries have their own big, intractable challenges. And quantum computers offer potential solutions. For instance, quantum computing can shed light on processes of molecular and chemical interactions," says L. Venkata Subramaniam, Senior Manager - AI Science and IBM Quantum Ambassador, IBM Research India. Mitsubishi Chemical, Keio University and IBM are working together to model and study the complex mechanism for lithium superoxide rearrangement, a key chemical step in lithium-oxygen batteries. Using a quantum computing setup enables them to more accurately measure the reactant energy used to produce electricity in a Li-air battery.

So, is quantum computing only for researchers and designers? No, every large business house can use it to optimise its running. "Quantum computing is enabling ExxonMobil to solve several computationally-

Quantum computer mixing chamber (Credit: IBM Research)

challenging problems such as optimising natural gas supply and the discovery of new materials for efficient carbon capture," says Subramaniam.

Efficient shipping of liquid natural gas (LNG) is extremely critical, because people could run out of power without it! The global LNG industry involves thousands of voyages across the world. Optimising these, at the most basic level, would involve millions of discrete decisions. If you take into consideration uncertainties like bad weather and market fluctuation, it could run into even trillions.

According to Dr Vijay Swarup, Director of Technology at ExxonMobil, this means that the number of combinations they would need to consider would be larger than the number of atoms in the entire universe. ExxonMobil is working on modelling maritime inventory routing on IBM's quantum setup.

Think quantum

"We tend to think of problems and solutions in classical ways, for which the existing classical computational ecosystem is fast and powerful enough. Most of the standard/classical problems can be adequately addressed within the existing infrastructure. However, there exists a certain class of problems that quantum computers can solve in more effective ways," explains Rahul Mahajan, CTO, Nagarro.

"Drug discovery, security, and search are a few domains where we foresee some interesting applications using quantum computers. Given a sufficiently large quantum computer, there exist a few algorithms like Grover's and Shor's, which can be deployed on quantum machines to execute the solution faster. For example, a class of RSA algorithms use factorisation based encryption, leveraging quantum parallelism in qubits registers, and simultaneous evaluation paths can be traversed," he says.

"Quantum computing makes different types of seemingly intractable optimisation problems, like routing of delivery vehicles, more manageable. Another class of problems comes from chemical reactions. These are by nature quantum mechanical. The best way to understand them is via simulations on a quantum computer. With enough qubits, we could tackle problems in electric battery design, find new molecules, and aid drug discovery. The societal benefits that would follow are immense," says Professor Anil Prabhakar, Department of Electrical Engineering, IIT Madras.

In the last few years, there have been significant developments in this space—mergers, alliances, stock launches and public-private partnerships; better algorithms, full-stack solutions, hardware upgrades and concrete tech roadmaps; and an expanding portfolio of applications. We see people rethinking quantum deployments, often opting for quantum simulators, and hybrid systems with quantum computers working hand-in-hand with CPUs and GPUs. We see the emergence of cloud ecosystems that mask the complexity, enabling even non-quantum experts in different fields to use quantum computing to solve stubborn problems.

We do not have to wait for quantum computing to replace classical systems, to finally agree that quantum is here—because that will probably never happen. We need to understand that classical and quantum